# Basics of DSA

Total points 24/25

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### 1. What is a Data Structure? * 1/1

- A) A programming language
- B) A way of organizing and storing data efficiently
- C) A compiler
- D) An operating system

### 2. Which of the following is NOT a real-life application of data structures? * 1/1

- A) Library Management
- B) Banking System
- C) Railway Reservation
- D) Photo Editing

### 3. Which of the following is a benefit of using data structures? * 1/1

- A) Slower execution
- B) Efficient storage ✓ (selected)
- C) Increased memory waste
- D) Difficult searching

### 4. Which data structure is classified as a Linear Data Structure? * 1/1

- A) Tree
- B) Graph
- C) Stack ✓ (selected)
- D) Heap

### 5. Which of the following is a Non-linear Data Structure? * 1/1

- A) Queue
- B) Array
- C) Linked List
- D) Tree ✓ (selected)

### 6. Linear data structures are arranged in: * 1/1

- A) Hierarchical order
- B) Circular order
- C) Sequential order
- D) Random order

### 7. Arrays store elements in: * 1/1

- A) Random memory locations
- B) Contiguous memory locations
- C) External memory
- D) Cache only

### 8. Which operation in an array generally requires shifting elements? * …/1

- A) Traversal
- B) Searching
- C) Insertion
- D) Printing

No correct answers

**9. What is the time complexity of inserting an element at the beginning of an array?** * 1/1

- A) O(1)
- B) O(log n)
- C) O(n) ✓
- D) $O(n^2)$

**10. Which is an advantage of arrays?** * 1/1

- A) Dynamic size
- B) Random access ✓
- C) No memory limitation
- D) Fast insertion

**11. Which is a disadvantage of arrays?** * 1/1

- A) Random access
- B) Fixed size ✓
- C) Cache friendly
- D) Easy traversal

### 12. In array deletion, elements are shifted: *

1/1

- A) Right
- B) Left
- C) Up
- D) Down

### 13. A linked list consists of: *

1/1

- A) Only data
- B) Data and index
- C) Data and pointer
- D) Only pointer

### 14. Linked list nodes are stored in: *

1/1

- A) Contiguous memory
- B) Non-contiguous memory
- C) ROM
- D) Cache memory

### 15. Which is an advantage of a linked list? * 1/1

- A) Random access
- B) Dynamic size ✓
- C) Fixed size
- D) Faster indexing

### 16. Which of the following is NOT an advantage of linked lists? * 1/1

- A) Dynamic memory allocation
- B) Easy insertion
- C) Random access ✓
- D) Easy deletion

### 17. Insertion at the beginning of a linked list takes: * 1/1

- A) O(n)
- B) O(log n)
- C) O(1) ✓
- D) $O(n^2)$

**18. Which data structure provides random access?** * 1/1

- A) Linked List
- B) Queue
- C) Array ✓
- D) Stack

19. The index of the first element in most programming languages is: * 1/1

- A. 0 ✓
- B. 1
- C. -1
- D. Depends on the compiler

20. Which operation is the fastest in an array? * 1/1

- A. Insertion at the beginning
- B. Deletion from the middle
- C. Accessing an element by index ✓
- D. Searching for an element in an unsorted array

21.Which part of a linked list node stores the address of the next node? * 1/1

- A. Data field
- B. Pointer field ✓ (selected)
- C. Index field
- D. Header field

22. Which linked list allows traversal in both directions? * 1/1

- A. Singly Linked List
- B. Circular Linked List
- C. Doubly Linked List ✓ (selected)
- D. Array

23. Which pointer points to the first node in a linked list? * 1/1

- A. Tail
- B. Head ✓ (selected)
- C. Root
- D. Front

24. What does the last node of a singly linked list point to? * 1/1

- ○ A. First node
- ○ B. Previous node
- ◉ C. NULL ✓
- ○ D. Head

25. Which data structure is better for frequent insertions and deletions? * 1/1

- ○ A. Array
- ◉ B. Linked List ✓
- ○ C. Stack
- ○ D. Queue